PHYS 4041/5041
Fall 2023
Homework 1
Due: In class, Friday September 8, 2023

1. Find the general solutions to the following:

(a) $y' = \frac{2xy^2+x}{x^2y-y}$

(b) $y' + y = \exp(x)$

(c) $y' = -y\tanh(x) + 2\exp(x)$

(d) $(2xe^{3y} + e^x)dx + (3x^2e^{3y} - y^2)dy = 0$

2. I like to think about radioactive decay in this way. Every atom of a given species must have the same probability of decaying within a given time interval. Thus, the decay rate of a sample of atoms must be proportional to the number of atoms in that sample. Consider the situation where we have three types of atoms which are related by the decay sequence $1 \to 2 \to 3$. Let N_1, N_2, and N_3 represent the number of atoms of each type. Let's also suppose that atom type 3 is stable so it does not decay. According to the above, we then have

$$\begin{aligned} \frac{dN_1}{dt} &= -\lambda_1 N_1 \\ \frac{dN_2}{dt} &= \lambda_1 N_1 - \lambda_2 N_2 \\ \frac{dN_3}{dt} &= \lambda_2 N_2, \end{aligned}$$

where λ_1 and λ_2 are the decay rate constants (related to the half life) for atom types 1 and 2, respectively. Note that atom type 1 only changes due to decays, atom type 2 changes due to feeding from atom type 1 and its decay, and atom type 3 changes only due to feeding from atom type 2. Assuming $N_1(0) = N_0$, $N_2(0) = 0$, and $N_3(0) = 0$, determine the number of all three atom types as a function of time. What is $N_1 + N_2 + N_3$?

3. Find the solution to

$$y'' - 3y' + 2y = 0,$$

subject to $y(0) = 0$ and $y'(0) = 1$.

4. Consider the following second-order homogeneous ODE with constant coefficients:

$$y'' + 2by' + (b^2 - \epsilon^2)y = 0,$$

where b and $\epsilon > 0$ are constants.

(a) Show that the general solution is

$$y = Ae^{(-b-\epsilon)x} + Be^{(-b+\epsilon)x},$$

where A and B are constants.

(b) This solution can also be written as

$$\begin{aligned} y &= e^{(-b-\epsilon)x}[A + Be^{2\epsilon x}] \\ &= e^{(-b-\epsilon)x}[A + B(1 + 2\epsilon x + \frac{1}{2}(2\epsilon x)^2 + \ldots)], \end{aligned}$$

where a Taylor series has been used in the second step. Defining the new constants $A' = A + B$ and $B' = 2\epsilon B$, we can now write

$$y = e^{(-b-\epsilon)x}[A' + B'x + B'\epsilon x^2 + \ldots].$$

Show that taking the limit $\epsilon \to 0$ (while keeping A' and B' constant) produces the expected result for the general solution. Thus, with some care, the general solution for case of degenerate roots can be deduced from the general solution for non-degenerate roots.

5. Consider the first-order differential equation

$$(6x^2 - y + 3)dx + (3y^2 - x - 2)dy = 0.$$

Determine the general solution.

6. Two of the named second-order homogeneous linear ordinary differential equations are

Legendre's Equation: $(1 - x^2)y'' - 2xy' + l(l + 1)y = 0$

Hypergeometric Equation: $x(x - 1)y'' + [(1 + a + b)x - c)]y' + aby = 0.$

Show that the substitutions

$$\begin{aligned} x &= \frac{1 - u}{2} \\ a &= -l \\ b &= l + 1 \\ c &= 1 \end{aligned}$$

convert the Hypergeometric Equation into Legendre's Equation.